

Cyber Security Report

Weeks 4 – 6

Internship Tasks and Security Implementation



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1. Introduction

This report summarizes the cybersecurity internship tasks completed during Weeks 4–6. The focus of these tasks was advanced threat detection, ethical hacking, API security hardening, penetration testing, and secure deployment practices. Multiple security tools and techniques were implemented to improve application security and identify vulnerabilities.

2. Week 4 Advanced Threat Detection & Web Security

2.1 Objectives

- Implement API security hardening.
- Configure security headers and CSP.
- Apply request rate limiting.
- Improve monitoring and logging.

2.2 Tasks Performed

Task	Implementation	Result
Rate Limiting	Used express-rate-limit middleware	Protected APIs from brute-force attacks
Security Headers	Configured Helmet middleware	Improved HTTP security
CORS Configuration	Restricted unauthorized origins	Prevented unsafe cross-origin access
CSP & HSTS	Configured CSP and HSTS headers	Reduced XSS and downgrade attack risks

2.3 Key Technologies

Express.js, Helmet, CORS, JWT Authentication, express-rate-limit, Winston Logger.

2.4 Screenshots & Evidence

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Noor Fatima>cd C:\Users\Noor Fatima\myapp

C:\Users\Noor Fatima\myapp>npm install express-rate-limit cors

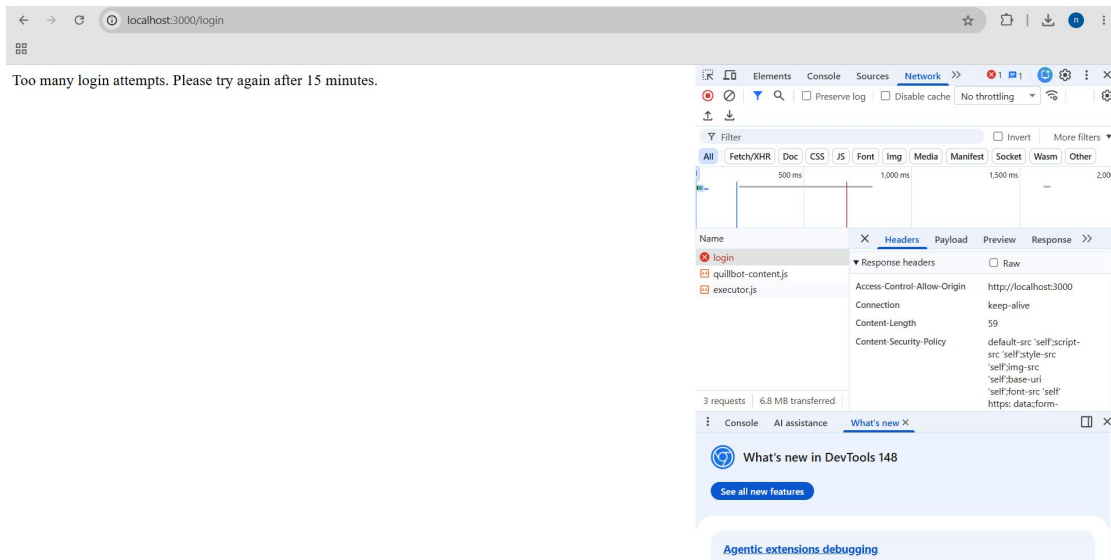
up to date, audited 112 packages in 5s

26 packages are looking for funding
  run `npm fund` for details

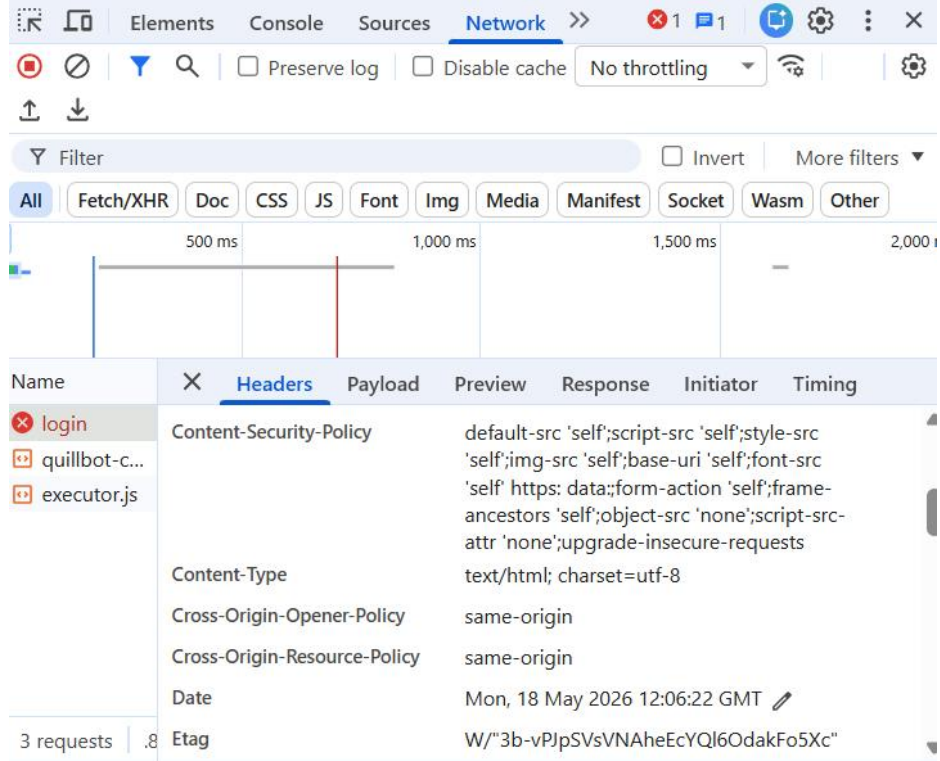
found 0 vulnerabilities

C:\Users\Noor Fatima\myapp>node app_4.js
2026-05-18T12:01:38.375Z [INFO]: Week 4 secure app started at http://localhost:3000
```

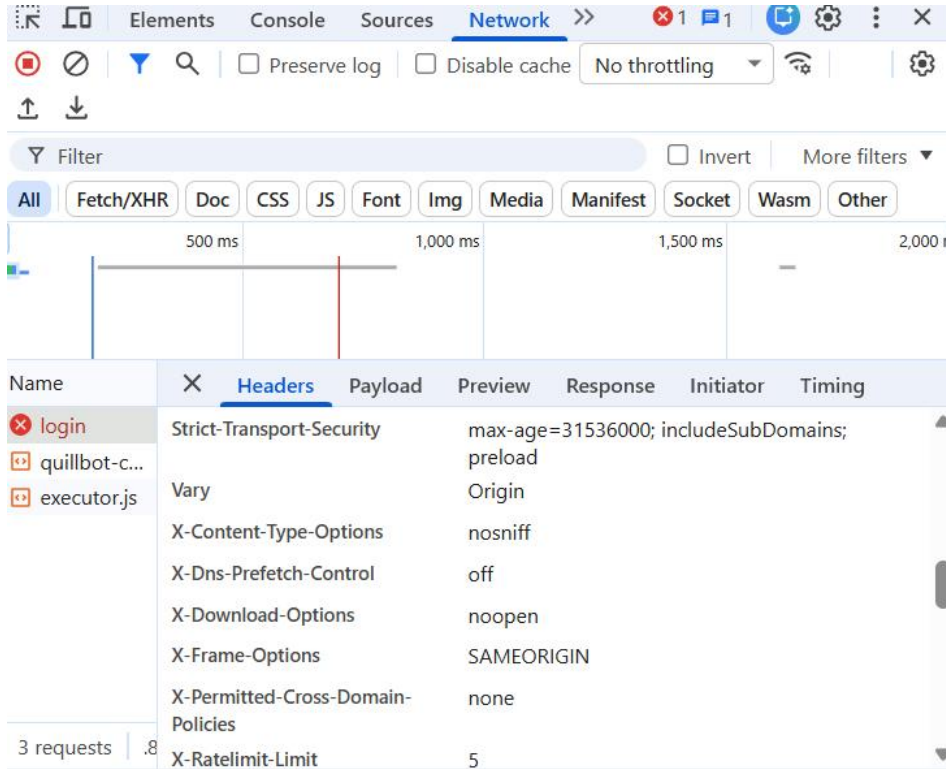
Screenshot 2026-05-18 170155.png



Screenshot 2026-05-18 170720.png



Screenshot 2026-05-18 170832.png



Screenshot 2026-05-18 170840.png

```
Command Prompt - node a x + v
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Noor Fatima>cd C:\Users\Noor Fatima\myapp
C:\Users\Noor Fatima\myapp>npm install csrf cookie-parser
up to date, audited 127 packages in 5s
26 packages are looking for funding
  run 'npm fund' for details
2 low severity vulnerabilities
To address all issues (including breaking changes), run:
  npm audit fix --force
Run 'npm audit' for details.
C:\Users\Noor Fatima\myapp>node app_5.js
2026-05-18T12:15:39.925Z [INFO]: Week 5 secure app started at http://localhost:3000
```

Screenshot 2026-05-18 171552.png

3. Week 5 Ethical Hacking & Exploiting Vulnerabilities

3.1 Objectives

- Perform ethical hacking in a test environment.
- Identify SQL Injection vulnerabilities.
- Implement CSRF protection.
- Perform penetration testing.

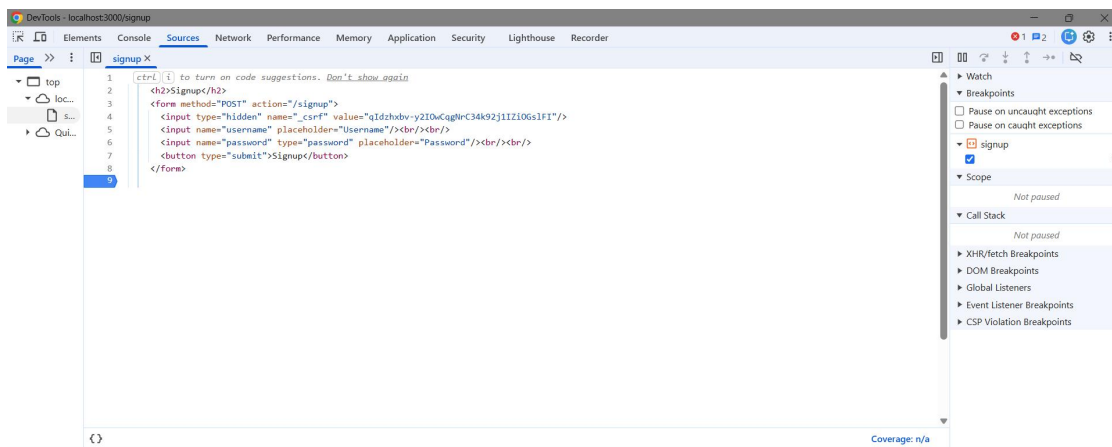
3.2 Security Testing Activities

Security Activity	Tool Used	Outcome
SQL Injection Testing	SQLMap	Detected and analyzed vulnerabilities
CSRF Testing	Burp Suite	Verified CSRF protection
Ethical Hacking	Kali Linux	Conducted controlled penetration testing
Secure Coding	Prepared Statements	Mitigated SQL injection risks

3.3 Implemented Security Fixes

- Prepared statements were used to prevent SQL Injection attacks.
- CSRF middleware was configured to validate requests.
- Input validation and sanitization were added.
- Authentication handling was improved.

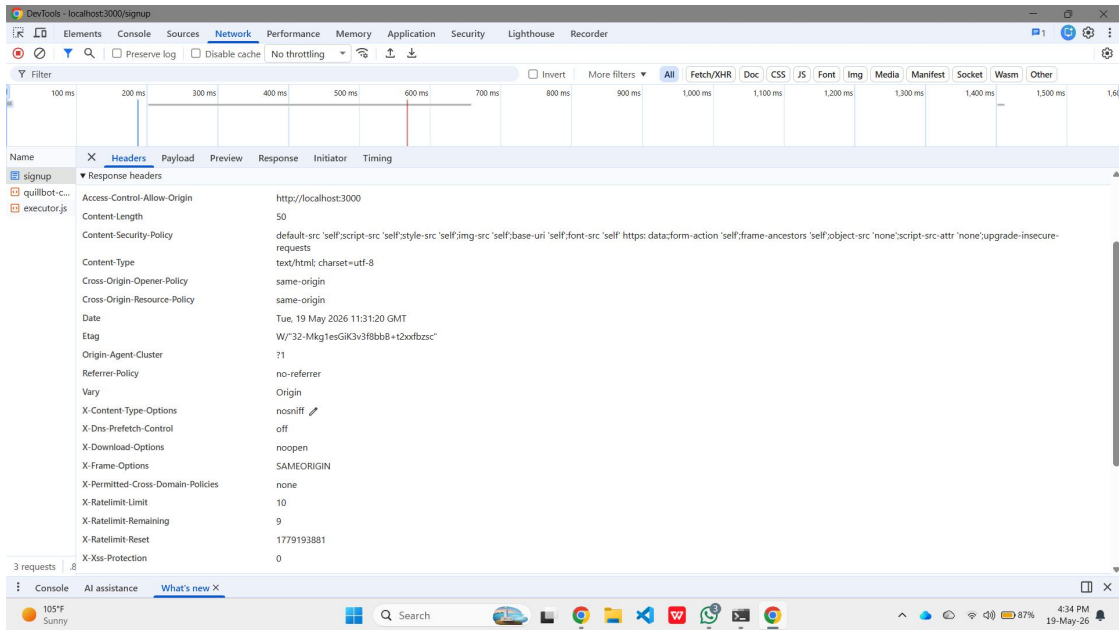
3.4 Screenshots & Testing Evidence



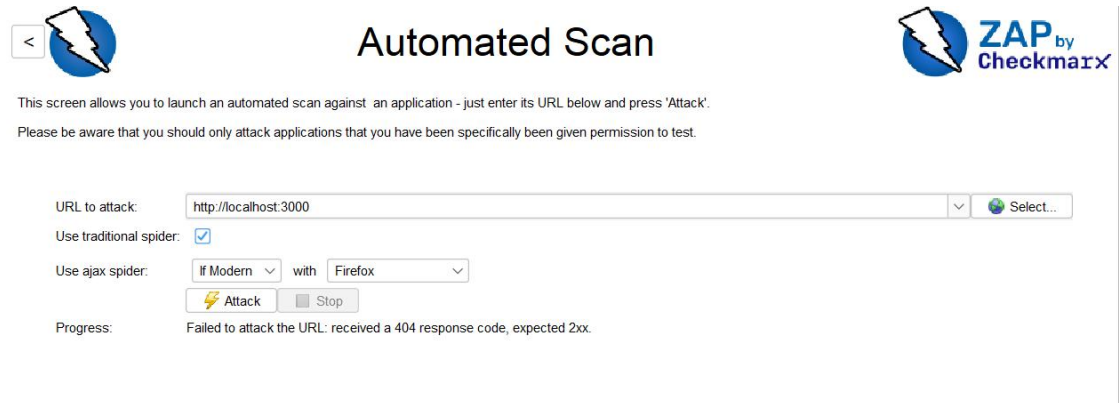
Screenshot 2026-05-19 163022.png



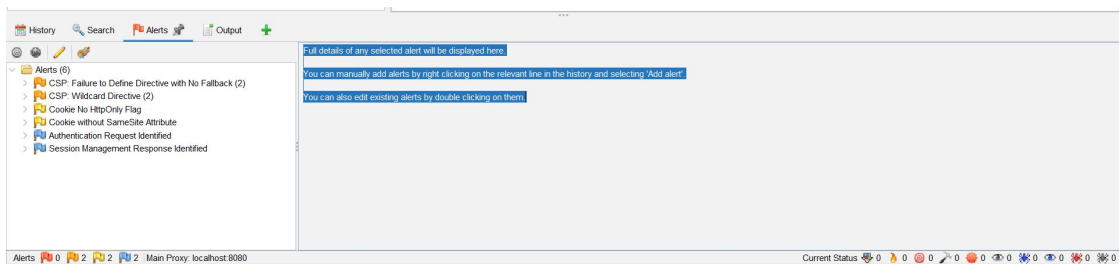
Screenshot 2026-05-19 163200.png



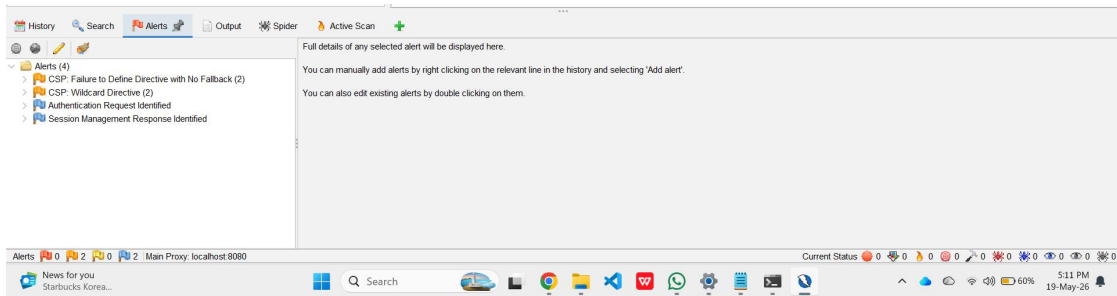
Screenshot 2026-05-19 163437.png



Screenshot 2026-05-19 164929.png



Screenshot 2026-05-19 165128.png



Screenshot 2026-05-19 171143.png

4. Week 6 Advanced Security Audits & Secure Deployment

4.1 Objectives

- Perform advanced security auditing.
- Review compliance with OWASP standards.
- Conduct penetration testing.
- Review secure deployment practices.

4.2 Security Audit Summary

Audit Tool	Purpose	Findings
OWASP ZAP	Web vulnerability scanning	Detected security weaknesses
Nikto	Server security testing	Reviewed server configuration
Lynis	System auditing	Checked system hardening
Burp Suite	Penetration testing	Validated web application security

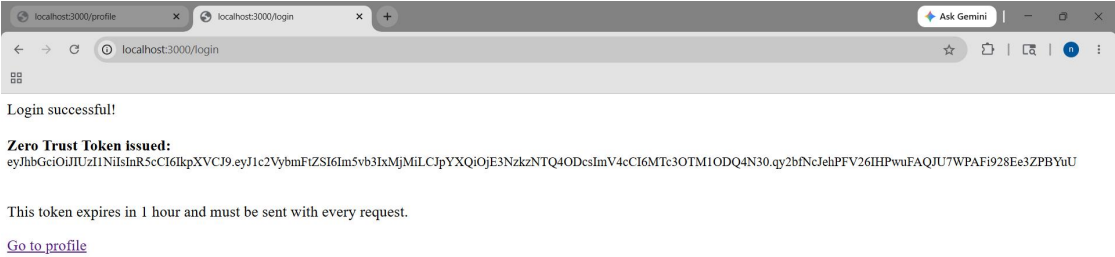
4.3 Secure Deployment Measures

- Reviewed dependency security and vulnerabilities.
- Implemented secure application configuration.
- Reviewed Docker security best practices.
- Performed penetration testing before deployment.

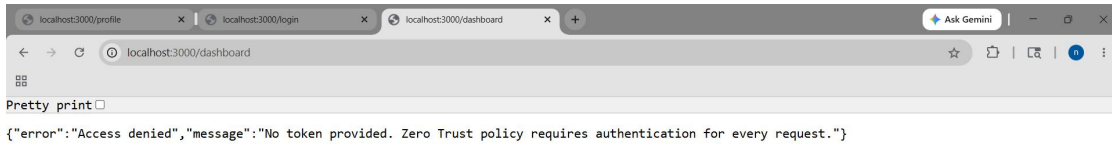
4.4 Screenshots & Audit Reports



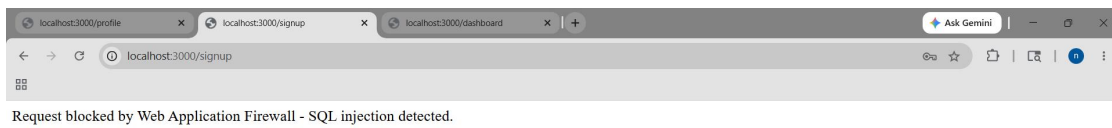
Screenshot 2026-05-21 141341.png



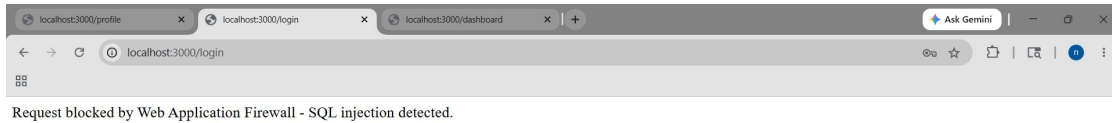
Screenshot 2026-05-21 141456.png



Screenshot 2026-05-21 141527.png



Screenshot 2026-05-21 142109.png



Screenshot 2026-05-21 142156.png

5. Bonus Tasks

Additional optional security tasks were completed to further improve application security and demonstrate advanced cybersecurity practices.

Bonus Task	Implementation	Purpose
Web Application Firewall	Implemented app_waf.js	Blocked suspicious requests
Advanced Security Controls	Implemented app_bonus.js	Improved application security
Zero Trust Concepts	Access control review	Enhanced authentication security

6. Tools & Technologies Used

- Node.js & Express.js
- Helmet
- Burp Suite
- OWASP ZAP
- SQLMap
- Kali Linux
- JWT Authentication
- CORS & CSP
- Winston Logger

7. Conclusion

The internship tasks completed during Weeks 4–6 provided practical exposure to web application security, ethical hacking, penetration testing, and security auditing. Multiple vulnerabilities were identified, tested, and mitigated using industry-standard security practices and tools. The project was successfully secured using authentication, rate limiting, CSP configuration, CSRF protection, and security auditing techniques.

8. GitHub & Documentation Notes

All implemented tasks, security fixes, and testing activities were documented and organized within the project repository. Supporting files included source code updates, security reports, screenshots, and audit evidence.